DBA5103

Operations Research and Analytics

Zhenyu Hu

Semester I, 2026/2027



NUS
National University
of Singapore

Course Information

- Lecture hours:
  - Wednesday 15:00–18:00, BIZ2-0509;
  - Friday 12:00-15:00, BIZ2-0509;
  - Friday 18:30-21:30, BIZ2-0509.
- Instructor: Hu Zhenyu
  - Email: bizhuz@nus.edu.sg
  - Personal website: https://sites.google.com/view/zhenyuhu/home
- Office hour: BIZ1 8-62, by appointment
- TA: Zhao Yipeng, e1521284@u.nus.edu
  Zhang Guoyun, e0773599@u.nus.edu
- Website: Canvas
  - Lecture materials
  - Assignments and final project

A bit more about myself

- B.S. in Mathematics
- Ph.D. in Industrial Engineering
- NUS Business School since 2015.
- On research:
  - Dynamic Programming
  - Game Theory and Mechanism Design
- On education:
  - https://www.youtube.com/watch?v=Q1IL-hXO27Q

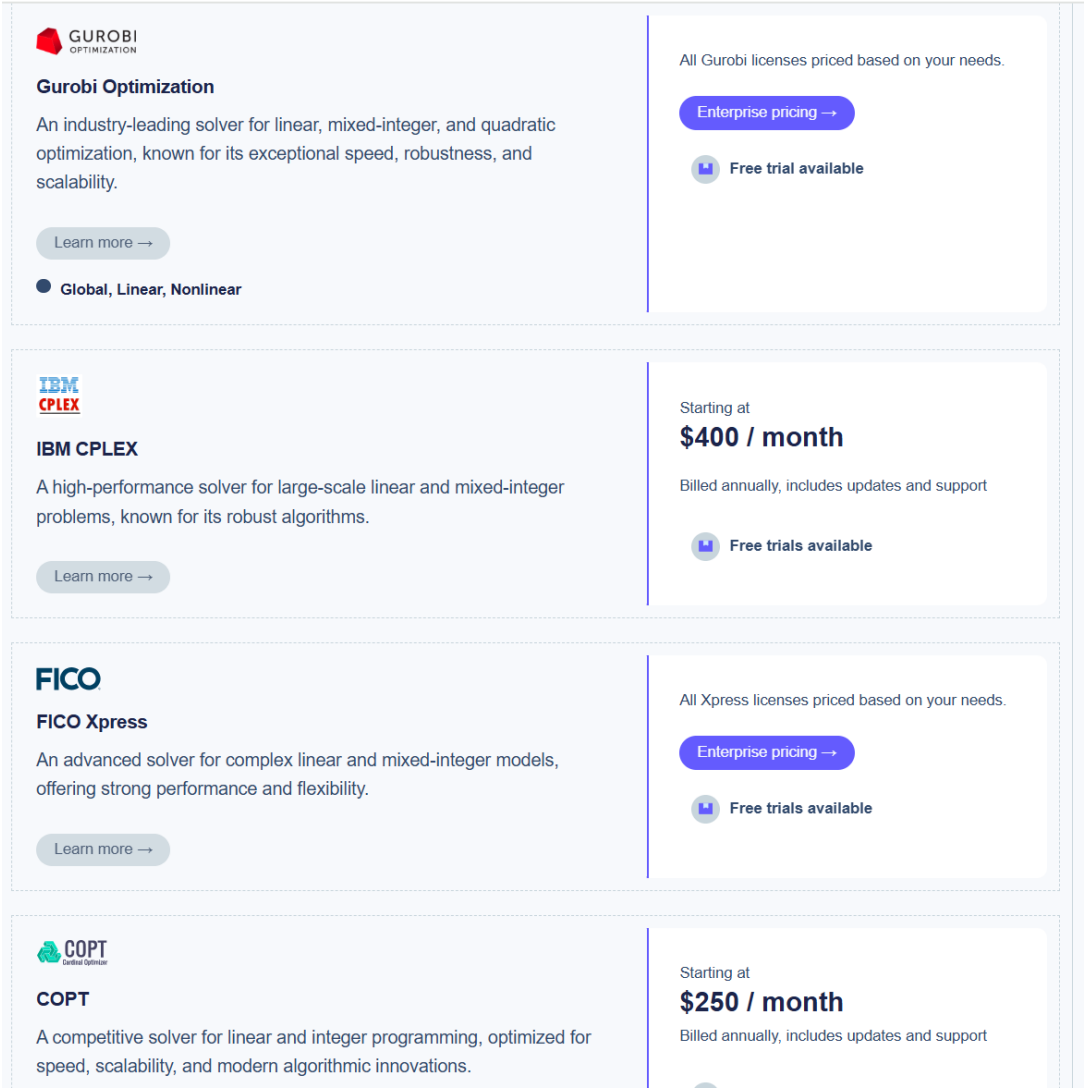
Textbook and Reference

- **Text:**
  - Introduction to Operations Research, Hillier and Lieberman (2015)
  - Applied Mathematical Programming, Bradley, Hax and Magnanti (1977)
  - Introduction to Linear Optimization, Bertsimas and Tsitsiklis (1997)
  - Mostly, lecture notes
- **Software:**
  - Python (Anaconda) and Gurobi
  - Excel Solver

Optimization Software

v · t · e	Mathematical optimization software [hide]
Data formats	LP · MPS · nl · OptML · OSiL · sol · xMPS
Modeling tools	AIMMS · AMPL · APMonitor · CMPL · CVX · CVXOPT · CVXPY · ECLiPSe-CLP · GEKKO · GAMS · GNU MathProg · JuMP · LINDO · OPL · MPL · OptimJ · PICOS · PuLP · Pyomo · ROML · TOMLAB · Xpress-Mosel · YALMIP · ZIMPL
LP, MILP* solvers	ABACUS* · APOPT* · Artelys Knitro* · BCP* · BDMLP · BPMPD · BPOPT · CLP · CBC* · CPLEX* · CSDP · DSDP · FortMP* · GCG* · GIPALS32 · GLPK/GLPSOL* · Gurobi* · HOPDM · LINDO* · Lp_solve · LOQO · MINOS · MINTO* · MOSEK* · OOPS · OOQP · PCx · QSopt · SAS/OR* · SCIP* · SoPlex · SOPT-IP* · Sulum Optimization Tools* · SYMPHONY* · XA* · Xpress-Optimizer*
QP, MIQP* solvers	APOPT* · Artelys Knitro* · BPMPD · BPOPT · BQPD · CBC* · CLP · CPLEX* · FortMP* · GloMIQO* · Gurobi* · IPOPT · LINDO* · LSSOL · LOQO · MINOS · MOSEK* · OOPS · OOQP · OSQP · QPOPT · QPSOL · SCIP* · XA Quadratic Solver · Xpress-Optimizer*
QCP, MIQCP* solvers	APOPT* · Artelys Knitro* · BPMPD · BPOPT · CPLEX* · GloMIQO* · Gurobi* · IPOPT · LINDO* · LOQO · MINOS · MOSEK* · SCIP* · Xpress-Optimizer* · Xpress-SLP*
SOCP, MISOCP* solvers	CPLEX* · DSDP · Gurobi* · LINDO* · LOQO · MOSEK* · SCIP* · SDPT3 · SeDuMi · Xpress-Optimizer*
SDP, MISDP* solvers	CSDP · DSDP · MOSEK · PENBMI · PENSDP · SCIP-SDP* · SDPA · SDPT3 · SeDuMi
NLP, MINLP* solvers	ALGENCAN · AlphaECP* · ANTIGONE* · AOA* · APOPT* · Artelys Knitro* · BARON* · Bonmin* · BPOPT · CONOPT · Couenne* · DICOPT* · FilMINT* · FilterSQP · Galahad library · ipfilter · IPOPT · LANCELOT · LINDO* · LOQO · LRAMBO · MIDACO* · MILANO* · MINLP BB* · MINOS · Minotaur* · MISQP* · NLPQLP · NPSOL · OQNLP* · PATHNLP · PENNON · SBB* · SCIP* · SNOPT* · SQPlab · WORHP · Xpress-SLP*
GO solvers	BARON · Couenne* · LINDO · SCIP
CP solvers	Artelys Kalis · Choco · Comet · CPLEX CP Optimizer · Gecode · Google CP Solver · JaCoP · OscaR
Metaheuristic solvers	OptaPlanner · LocalSolver

Optimization Software



https://ampl.com/pricing/

What You Should Expect

- **Mathematical Thinking**
  - Understand the basic logic behind algorithms
  - Understand the general principles in optimization
  - Judgement on the difficulty of a problem; be able to assess the quality of a solution

- **Analytics Skills**
  - Building up models
  - Picking the right tools
  - Implementing in Software

What I Expect

- **Prerequisite**
  - Linear algebra
  - Basic probability theory

- **Assessment**
  - Team Term Paper 30%
  - Individual Assignments 30%
  - In-Class Quiz 30%
  - Class Participation 10%

Homework 0

- Download and install python (Anaconda distribution) and gurobi (academic license)
  - Contact TA for help
- Form a team
  - 5-6 students per team
  - Deadline: recess week

Course Content

- Theory and application of linear programming
- Network flow models
- Integer programming
- Dynamic programming*

Operations Research in Practice

In a Nutshell

- **Decision**
- **Maximize/Minimize …**
- **Subject to …**

Areas of Operations Research

- Manufacturing
- Transportation
- Airline, Hotel and Retail

Synonyms and overlaps: Management Science, Decision Science, Industrial Engineering, Operations Management, Economics, Computer Science

Areas of Operations Research



Areas of Operations Research















































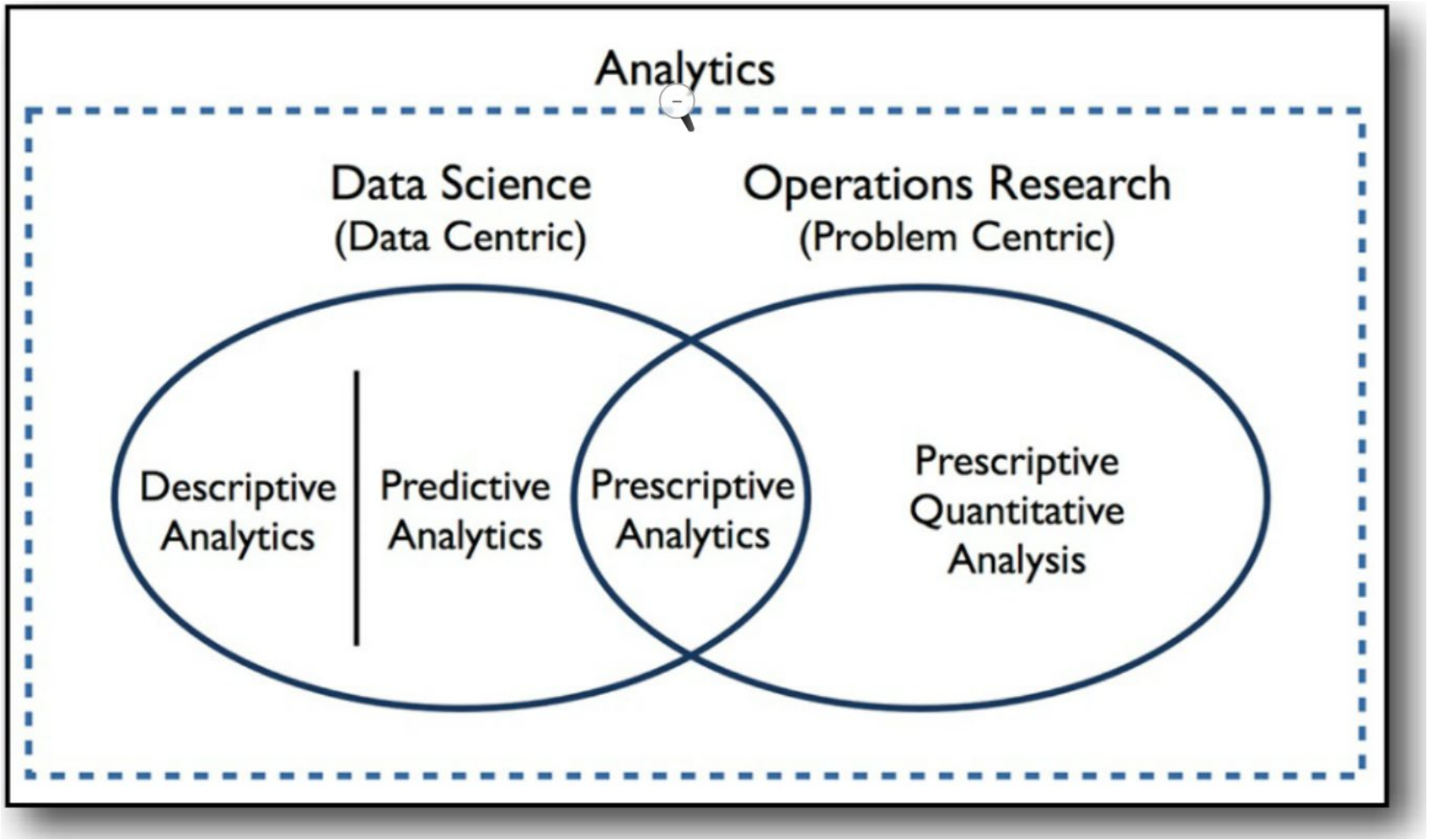








Rise of Analytics



Source: Robert Rose, "Defining analytics: a conceptual framework", 2016, *OR/MS Today,* Vol. 43, No.3

Ford

MAR 17, 2011

FORD OPENS FLEXIBLE, GREEN MICHIGAN ASSEMBLY PLANT WITH PRODUCTION OF ALL-NEW FORD FOCUS

- Transformed Michigan Assembly Plant (MAP), which will produce the fuel-efficient new global Ford Focus for North American customers, features flexible manufacturing, environmentally friendly practices and a highly trained work force
- A $550 million investment transformed the plant, creating new benchmarks for flexible manufacturing. Multiple models to run down the same production line, making Michigan Assembly the world's first plant to build gasoline-powered, battery electric, hybrid electric and plug-in hybrid electric vehicles on the same line
- MAP also features one of Michigan's largest solar-powered generation systems and electric vehicle charging stations

Ford Motor Company celebrated the launch of the 2012 Ford Focus with employees, dealers, suppliers, media and other invited guests at the Michigan Assembly Plant.

Click here to download related images.

Flexible Manufacturing



FOCUS

Starting MSRP $17,225[1]



MUSTANG

Starting MSRP $24,645[1]



C-MAX

Starting MSRP $24,170[1]

United States - Chicago Assembly Plant +

United States - Kentucky Truck Plant +

Unites States - Michigan Assembly Plant +

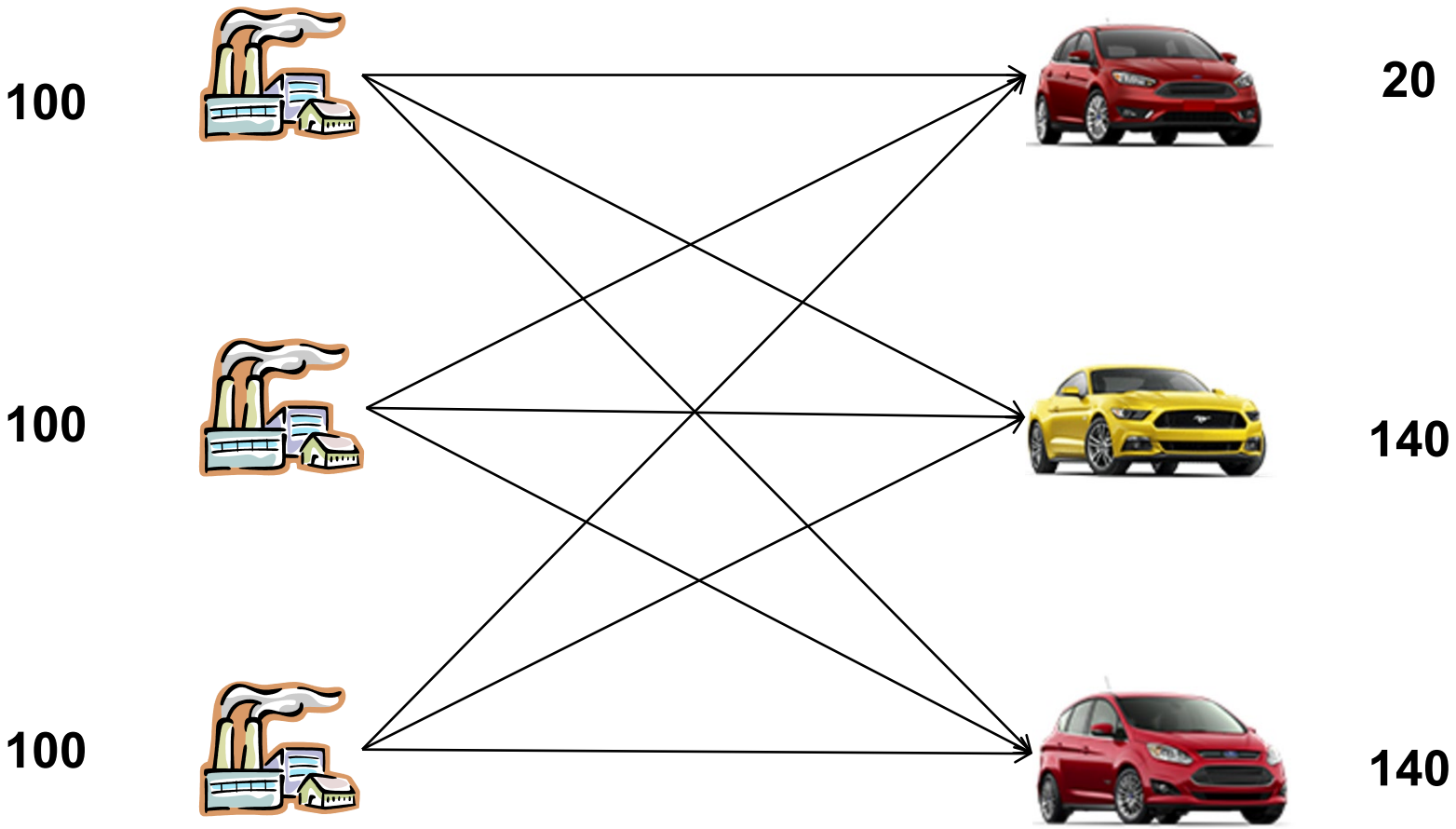
Traditional Dedicated System

100   20

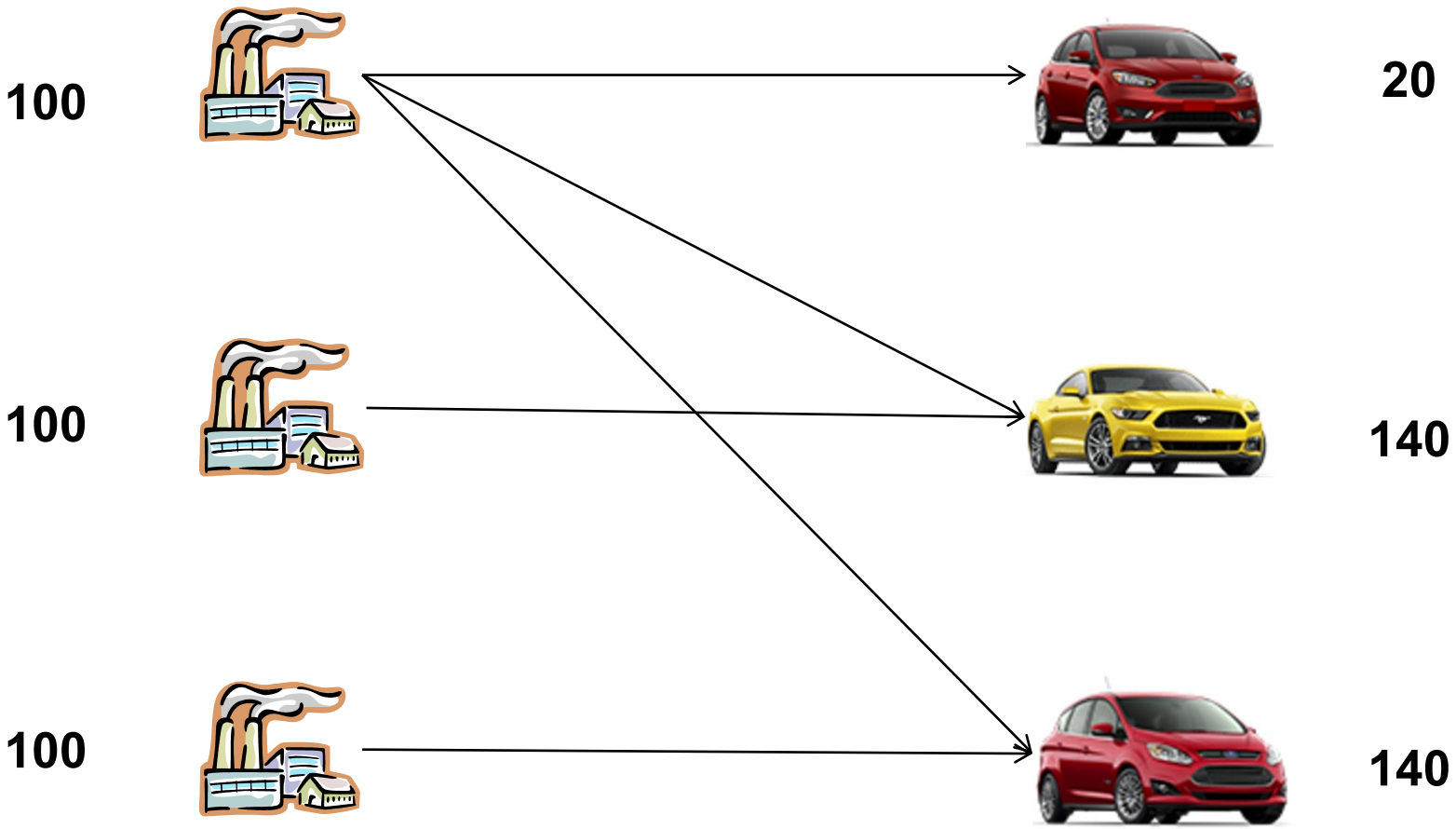
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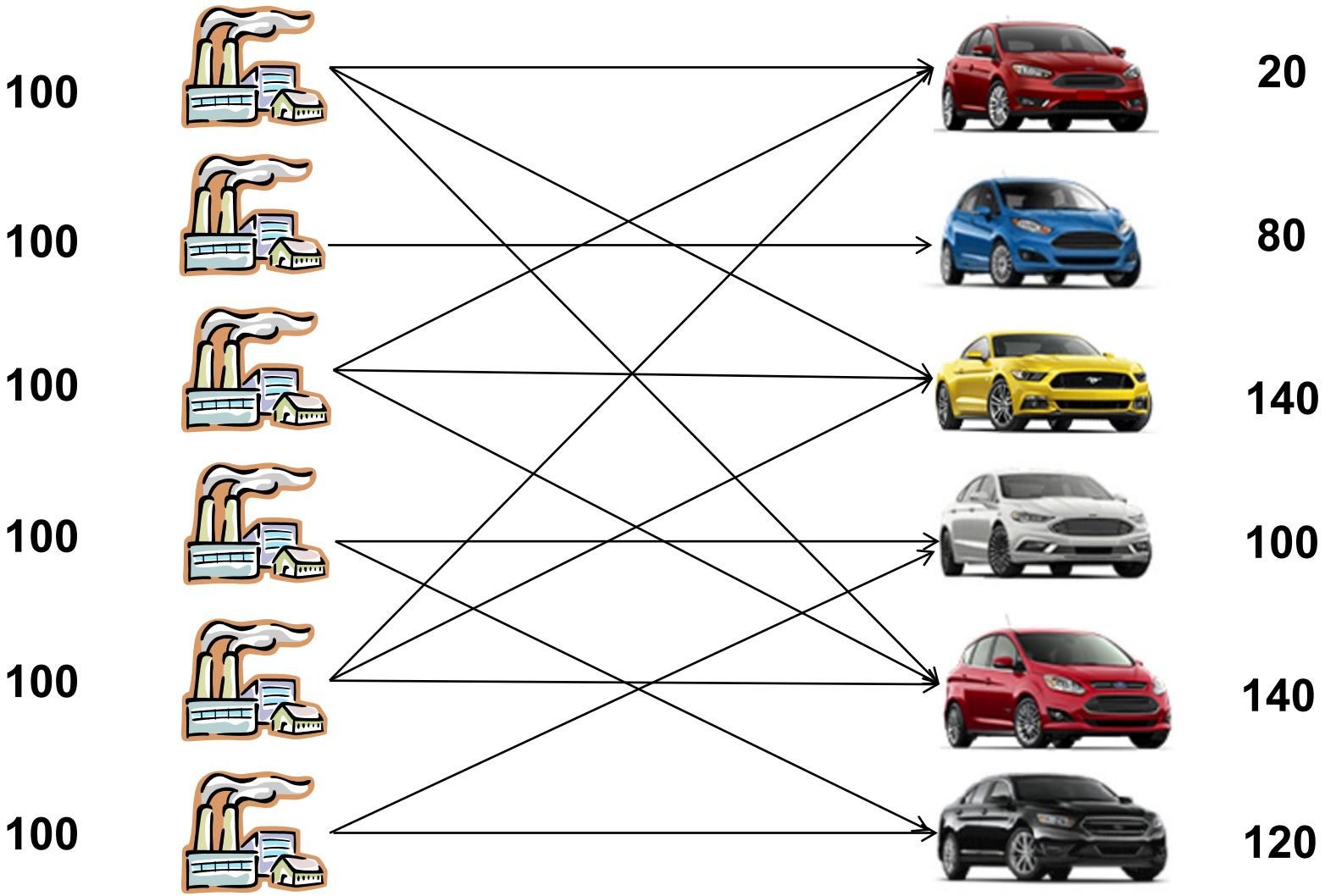
Full Flexible System



If we know demand…



How to Compute the Maximum Sales?



Operations Research Tools

- Given a design, how do we evaluate its performance?
  - Linear programming
  - Maximum flow
  - Simulation

Uber

Grab





滴滴

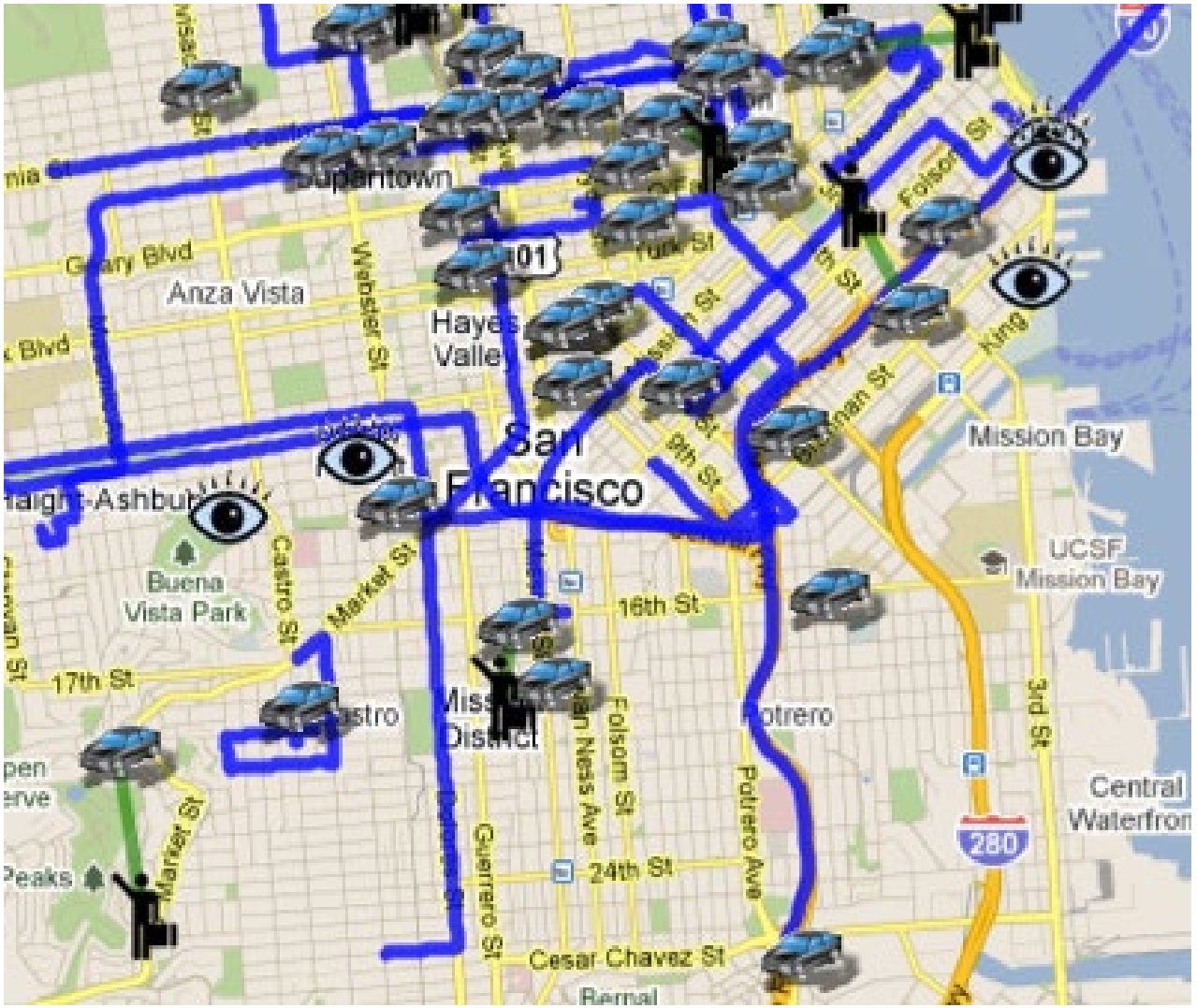
滴滴一下 美好出行

Uber/Grab

- What does Uber/Grab offer?

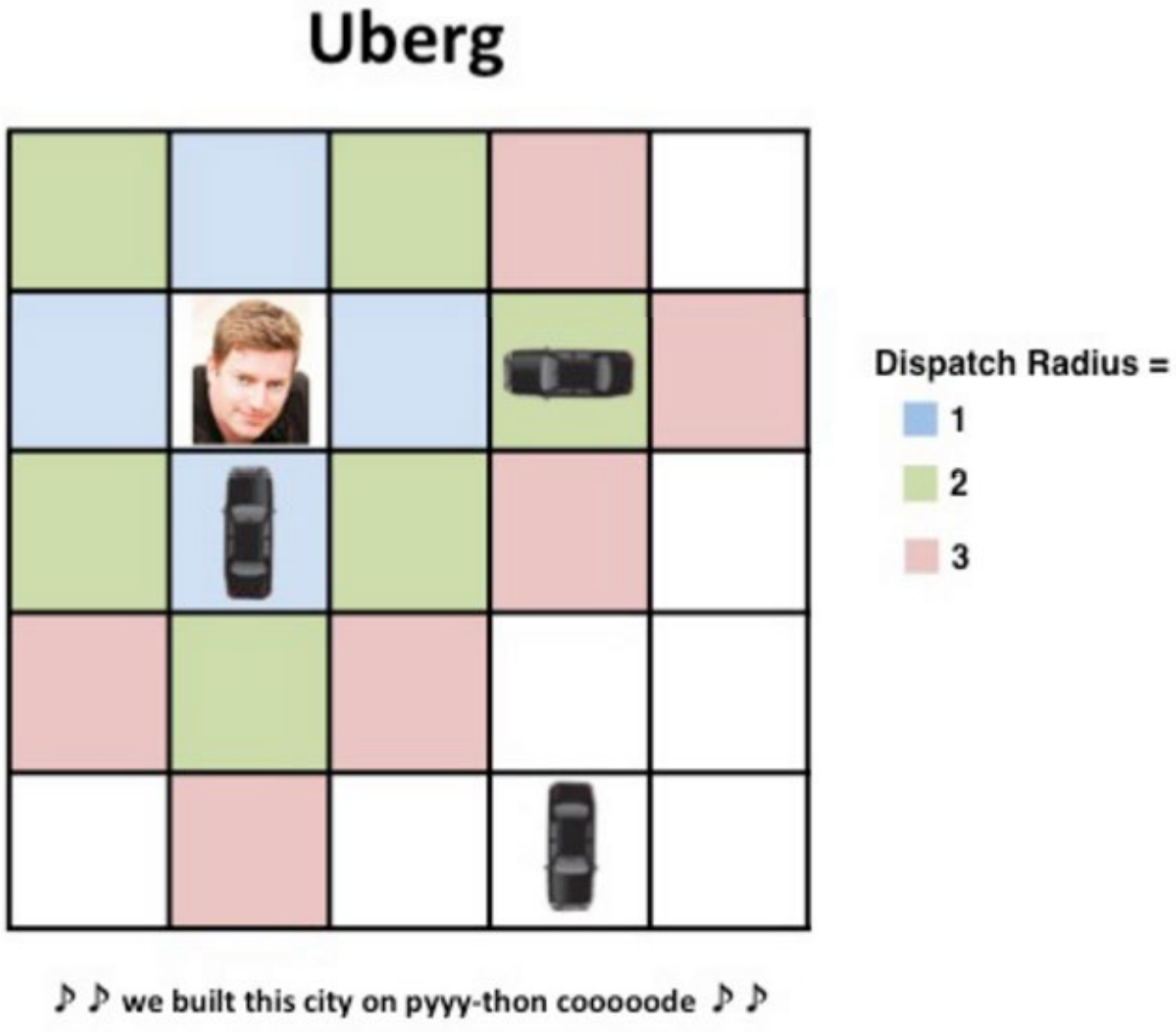
- Matching Algorithms
  - Dispatch system
  - Surge pricing
  - Uber POOL/Grab Share

Dispatch System



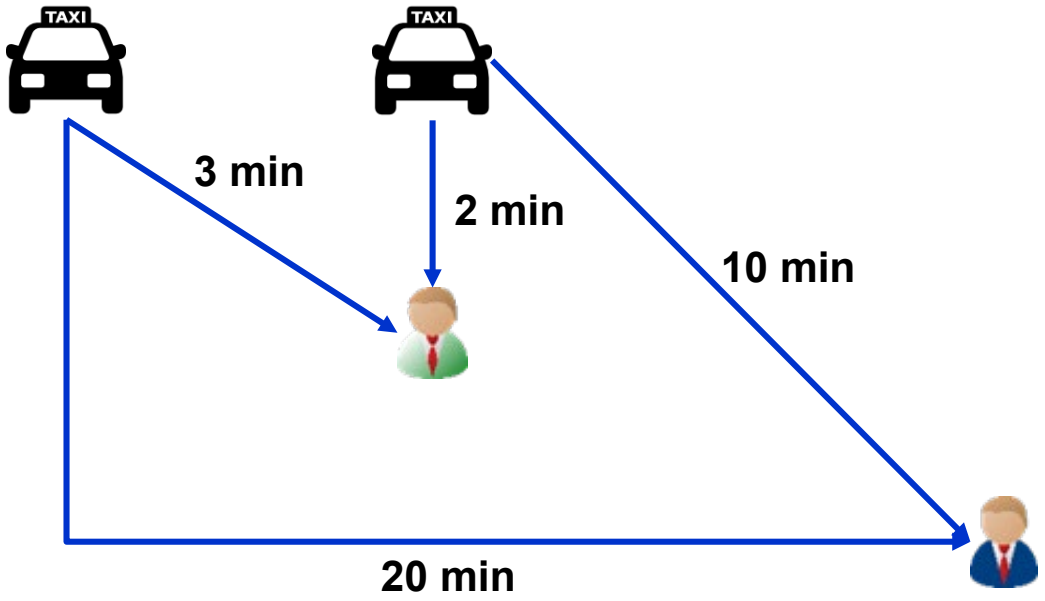
Source: http://www.wired.com/2011/04/app-stars-uber/

Greedy Approach



Source: https://newsroom.uber.com/semi-automated-science-using-an-ai-simulation-framework/

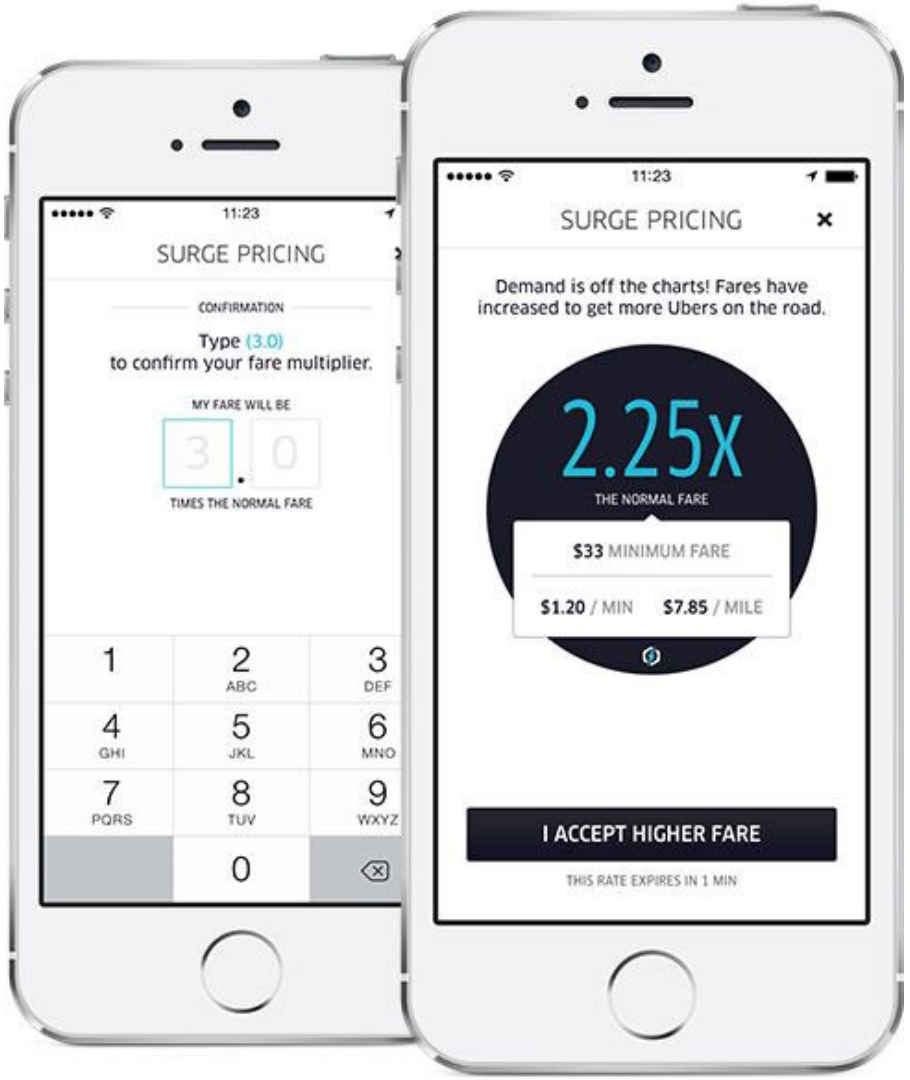
Is Greedy Good?



Operations Research Tools

- Linear programming.
- Shortest Path.
- Assignment Problem.

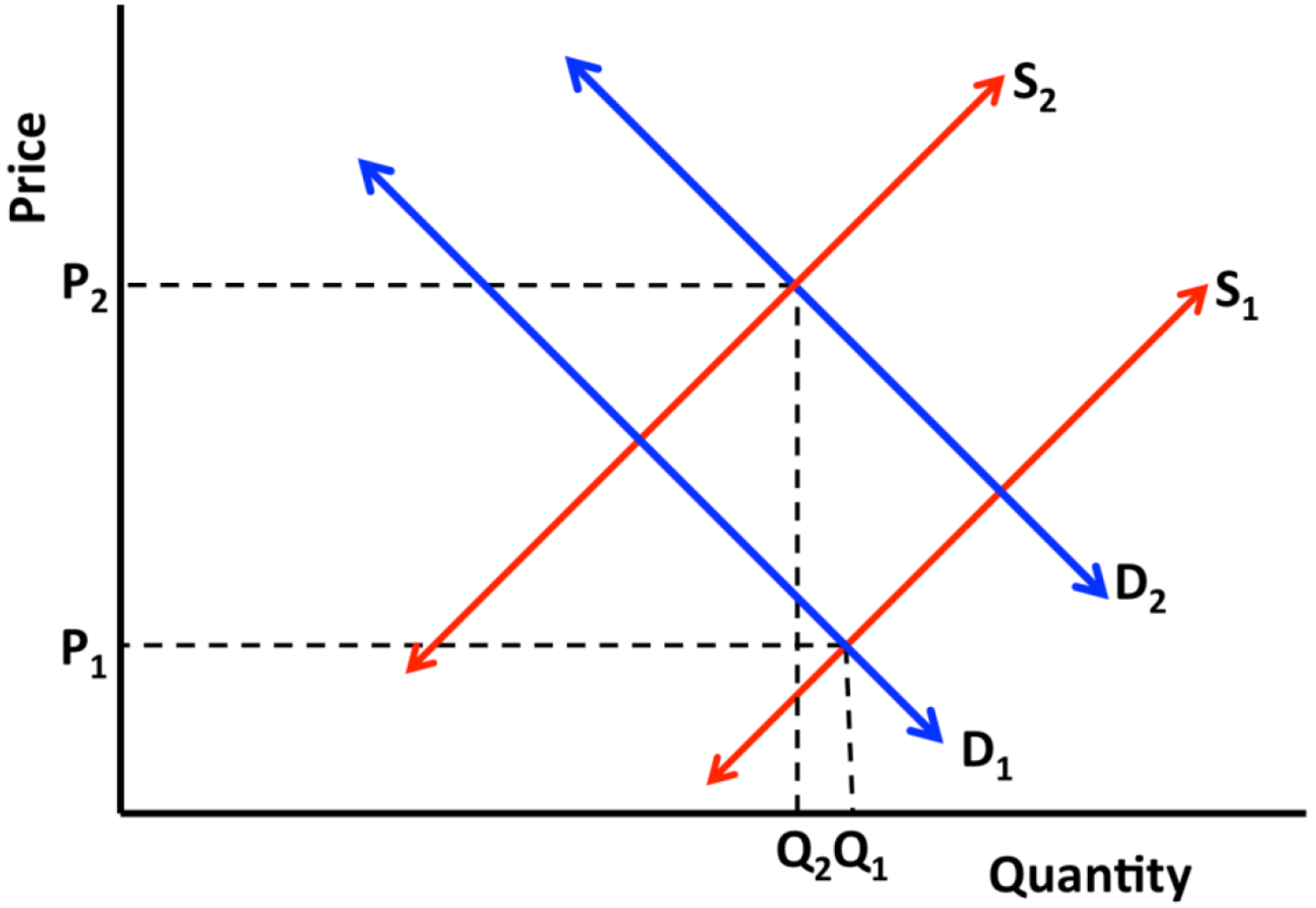
Surge Pricing



Why?

- **Profit Maximization?**
  - **Compare to airline industry**

ECON 1001



Source: https://newsroom.uber.com/guest-post-a-deeper-look-at-ubers-dynamic-pricing-model/

A Model for Supply and Demand

- ***n*** **locations.**
- **Demand at location *i*:** $a_i - b_i p_i$
- **Supply at location *i*:**

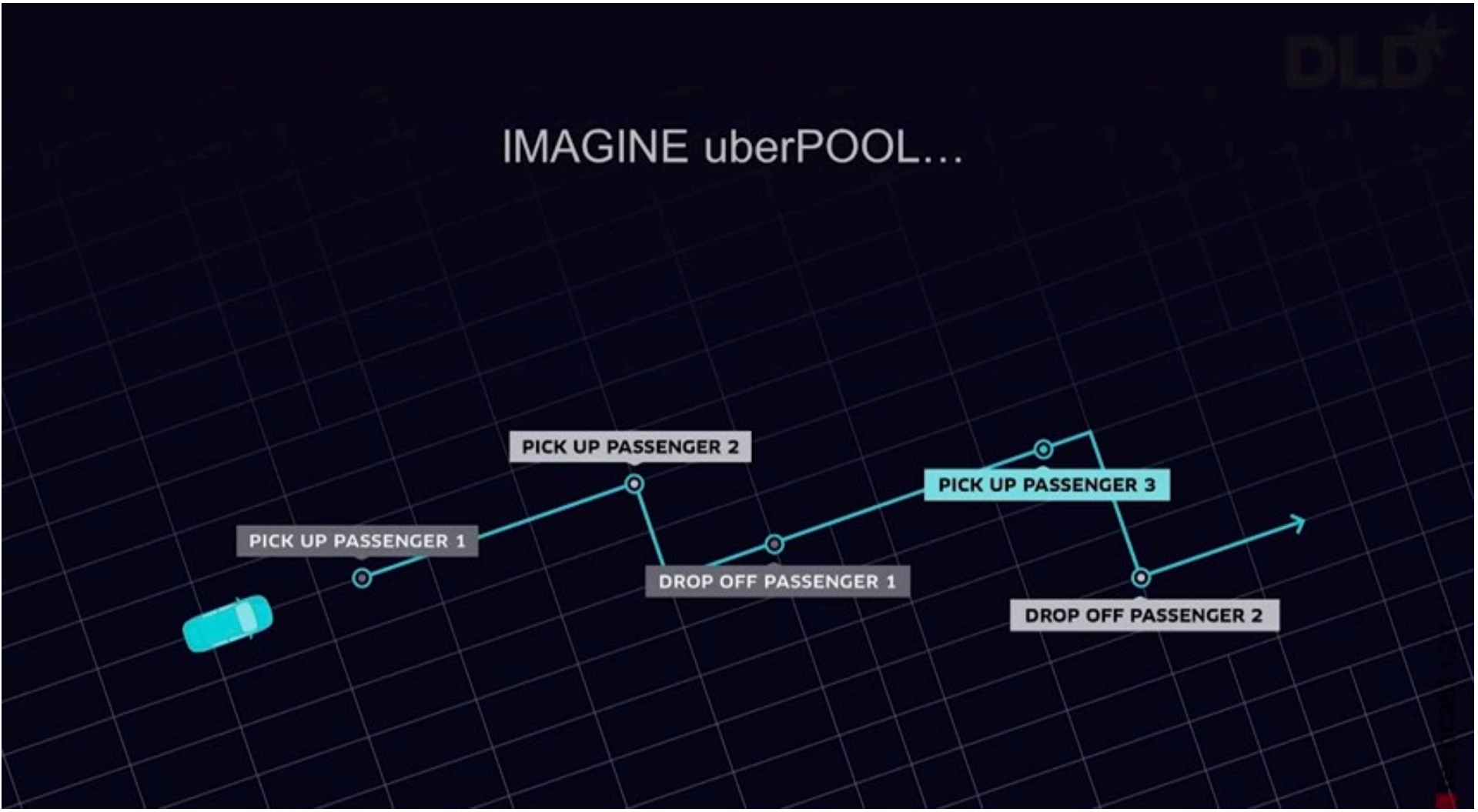
$$c_i + d_i p_i - \sum_{j \neq i} \theta_{ij} p_j$$

Optimization Problem

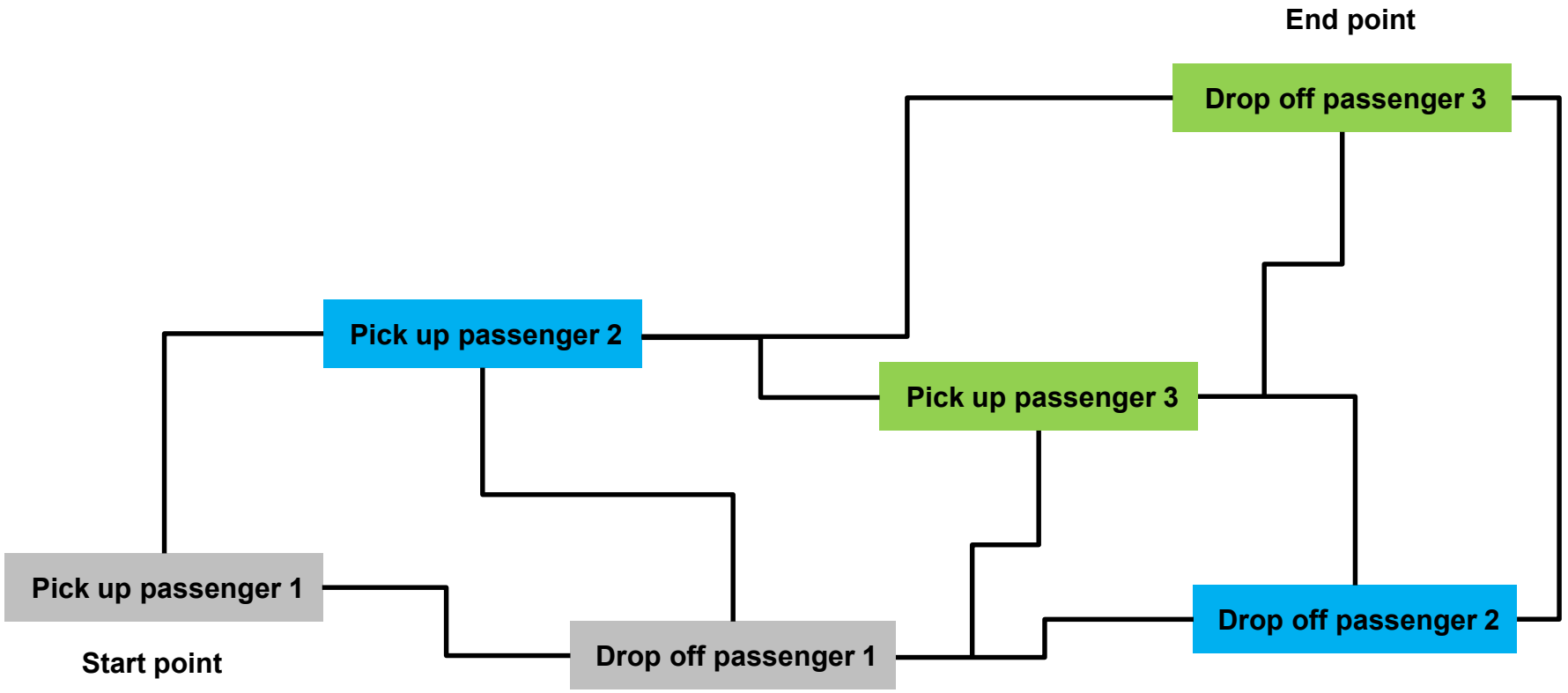
- **Non-profit**
  - Linear programming (in fact, linear algebra)

- **Profit**
  - Nonlinear programming

UberPOOL


IMAGINE uberPOOL…
PICK UP PASSENGER 1
PICK UP PASSENGER 2
DROP OFF PASSENGER 1
PICK UP PASSENGER 3
DROP OFF PASSENGER 2

Traveling Salesman Problem



Solving TSP

- Linear programming.
- Integer programming.
- Heuristic design.

Rue La La



LAGOS ▸

Closing in 1 day, 07:01:02



First Look: Cool-Weather Style for Men ▸

Closing in 3 days, 07:01:02



Feel Brand New: Crisp Looks for Work ▸

Closing in 1 day, 07:01:02

Source: https://www.ruelala.com/boutique/

Rue La La



John Varvatos Star U.S.A. Linen Moto Jacket
$169.99 ~~$498.00~~
2 LEFT



Feldton Japanese Selvedge Jacket
$199.99 ~~$398.00~~

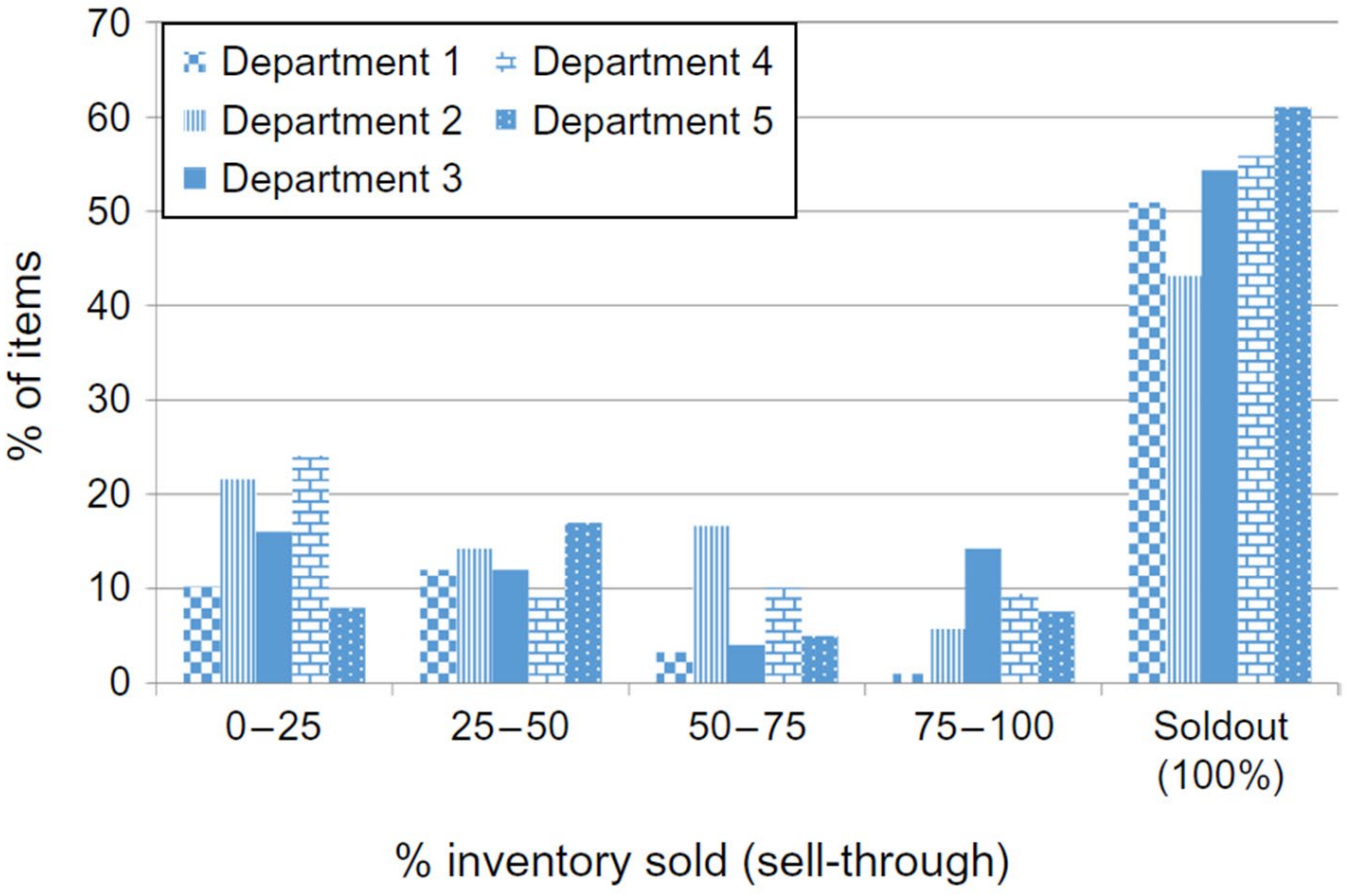


John Varvatos Star U.S.A. Linen-Blend Hooded Pullover
$84.99 ~~$248.00~~

Source: https://www.ruelala.com/boutique/

- **Legacy pricing process**
  - **Percentage markup on cost**
  - **Competitor's price**
  - **Subjective judgement**

Rue La La



Source: Ferreria, Lee, Simchi-Levi (2016), "Analytics for an Online Retailer: Demand Forecasting and Price Optimization", 2016, *MSOM,* Vol. 18, No.1

Price Optimization

- **Methodology**
  - **Integer programming/ linear programming**
- **Experiment**

Category	Estimate of percent increase in revenue (%)	Estimate of 90% confidence interval (%)	Estimate of 95% confidence interval (%)
A	−3.4	[−11.5, 7.7]	[−13.5, 10.1]
B	11.4	[3.9, 19.2]	[1.1, 21.0]
C	12.5	[1.1, 23.4]	[−2.0, 26.6]
D	13.7	[3.4, 22.8]	[0.0, 25.2]
E	23.8	[5.4, 47.6]	[0.0, 56.7]
Overall	9.7	[2.3, 17.8]	[0.0, 20.2]

Source: Ferreria, Lee, Simchi-Levi (2016), "Analytics for an Online Retailer: Demand Forecasting and Price Optimization", 2016, *MSOM,* Vol. 18, No.1

Other Ways of Improving Profit?

- **Markdown Management/Dynamic Pricing**
  - Inventory is determined before the selling season begins
  - Dynamically reduce price over the selling season to boost demand and increase revenue
- **When to decrease and by how much? Tradeoff?**

Dynamic Pricing

- Linear programming.
- Dynamic programming.